

CHAPTER – I

JAVA INTRODUCTION

PROGRAMMING LANGUAGE:

- A programming language is used to give the operations to the computer.
- It acts as a interface between human and the computer.
- Computers only understand the binary language (0s and 1s).
- Programming languages are human understandable language that converts the programming code into byte code.

TYPES OF PROGRAMMING LANGUAGE:

- **Low-level language-** hard to understand by the human (eg: binary language, assembly language).
- **High-level language-** human understandable language (eg: python, java, c++ etc).

JAVA:

- Java is an high level, object oriented and class based programming language.
- It is a platform independent language.
- Java was developed by sun microsystems in 1995.
- It is used to develop various applications.

ADVANTAGES OF JAVA:

- Platform independent
- Security
- Versatility
- Oops based
- Stability
- Huge libraries

COMPILER IN JAVA-

- Used to translate a java source code into byte code.
- Uses of compiler- checks syntax error, converts the java code into byte code.

INTERPRETER IN JAVA:

- It is used to executes the code line by line.
- Interpreter is the part of JVM.
- Uses of interpreter- runs byte code on various platforms, WORA.

JVM- JAVA VIRTUAL MACHINE:

- JVM is a engine that runs the program.
- It loads the code- verifies the code-executes code- manage memory.

JRE- JAVA RUNTIME ENVIRONMENT:

- It is package that contain everything needs to run a program.
- It includes JVM and libraries.
- It provides the environment where Java applications run.

JDK- JAVA DEVELOPMENT KIT:

- JDK provides a full software package to develop a java application.
- It gives a tools to write, run, debug and compile a code.
- JDK contains JVM and JRE.

IDE- INTERGRATED DEVELOPMENT ENVIRONMENT:

- IDE is programmers workspaces.
- Example: eclipse, VS code, Intelli J ect,.

JAVA MAGIC CODE:

- Every java .class files contains the magic code to identify.
- Magic code- CAFEBABE.

JIT- JUST -IN -TIME COMPILER OR INTERPRETER:

- JIT is used to run the java code faster than usual.
- It is a part of JVM.